

Clinical Risk Assessment Report: Patient 59

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 31.5% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.42x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +6.4%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.32 [Bottom 20%] (+6.4%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 0.72 [Top 13%] (+4.9%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 1.0 [Top 14%] (+2.8%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.77 [Top 22%] (+2.4%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.22 [Bottom 15%] (+1.7%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

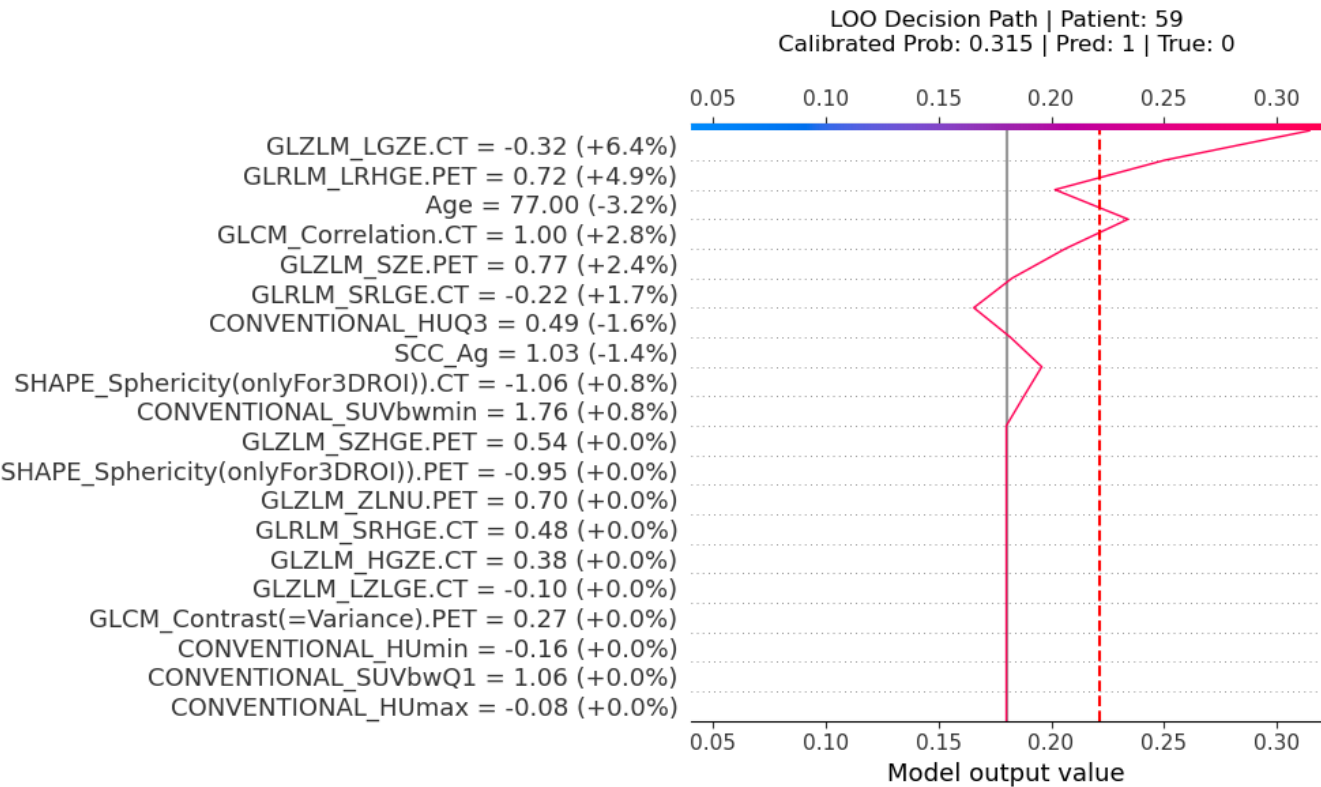
3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 77.0 [Top 6%] (-3.2%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.49 [Avg (58th %ile)] (-1.6%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 1.03 [Top 13%] (-1.4%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

